



COO/Palomar Observatory



Cover Sheet - Semester 2018A

200-inch Telescope

Instructions: Fill in the required information. When finished, save the form on your computer as a pdf document to be merged with the discussion section of your proposal.

Principal Investigator		Lead Observer (if not PI)	
Name:	Zheng Zheng		
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Office phone:			

The PI or the Lead Observer will be the single-point contact with the Palomar staff for all administrative and logistical matters pertaining to the project.

Program Title

Virgo3D: Dynamics and Stellar Populations of Galaxies in the Virgo Cluster

Abstract of Program Abstract must fit within the space provided. *Unformatted text only.*

Understanding the quenching of star formation is key to revealing the physical processes involved in galaxy evolution and transformation. Whether quenching is largely due to internal processes within galaxies or external processes due to the environment, and how these processes depend on the galaxy mass, are still open questions. These questions have been addressed through the study of the dynamics and stellar populations of relatively massive early type galaxies ($M_* > 10^9 M_\odot$) or the study of the Local Group dwarfs ($M_* < 10^7 M_\odot$), which has resulted in a morphology and mass gap in our understanding of these quenching processes, and about galaxy evolution in a more general context. Based on the deep Next Generation Virgo Cluster Survey (NGVS) imaging, we are conducting a 2-D spectroscopic survey of Virgo cluster galaxies that, for the first time,

Note: The abstract may be used for Educational Outreach purposes.

Institution	NAOC	Status ¹	Faculty	Priority	Principal
Linked to Keck ¹	No	Linked to 60-in ¹	No	Linked to Oschin ¹	No
Thesis Project	No	ToO Interrupts ¹	No	Large Project ¹	No
Date-Specific ²	No	Partial Nights ²	No	New/Continuing Project ³	New

¹Caltech observers only

²Provide full details in the discussion section. Download the [DS Supplement](#), fill in the required information and include it as an addendum to your proposal.

³For a new project, include in the discussion section an estimate of the total observing time needed for completion. For a continuation of an existing project, give a review of its status and an estimate of the observing time needed for completion.

Collaborators (Names and e-mail addresses)

Eric Peng <peng@pku.edu.cn>
 Elisa Toloba <etoloba@pacific.edu>
 Kyle Westfall <westfall@ucolick.org>
 Chengze Liu <cqliu@sjtu.edu.cn>
 Sungsoo Lim <slim@pku.edu.cn>

Observing Team (Observers expected to be present at Palomar)

Zheng Zheng
 Eric Peng

Observing Runs & Instruments Fill in the table below to specify the instruments requested and your acceptable observing runs.

- Select the **instrument(s)** you are requesting for the block from the pull-down menus. Double-instrument set-ups should not be requested unless both instruments are to be used during the scheduled block.
- Enter the **number of nights** you are requesting for the block.
- Specify your observing preferences as **P** = Preferred or **A** = Acceptable
- Use multiple request lines if you want your observing time split into two or more blocks (one line for each block).

L = Light (<28% moon-free)	D = Dark (>72% moon-free)
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DE = Grey Dark Early (28%-72% moon-free)

DL = Grey Dark Late (28%-72% moon-free)

Observing Time Preference & Instrument Selection 2018A													
		From	01 Feb	04 Feb	09 Feb	20 Feb	24 Feb	06 Mar	10 Mar	21 Mar	25 Mar	04 Apr	
		To	03 Feb	08 Feb	19 Feb	23 Feb	05 Mar	09 Mar	20 Mar	24 Mar	03 Apr	07 Apr	
		Approx RA @ Midnight	09h 00m - 10h 25m				10h 30m - 12h 20m				12h 25m - 13h 15m		
Block	Cass#	Prime#	Nights	1L	1DE	1D	1DL	2L	2DE	2D	2DL	3L	3DE
Blk 1	CWI	None	5							P			
Blk 2	None	None											
Blk 3	None	None											
Blk 4	None	None											
Blk 5	None	None											
Blk 6	None	None											

#Two-instrument requests must be fully justified, in that both instruments must be *integral* to the project.

*For *Own Equip*, provide name (below) and full description in proposal *Discussion*.

From	08 Apr	19 Apr	23 Apr	04 May	08 May	18 May	22 May	03 Jun	08 Jun	17 Jun	21 Jun	03 Jul	08 Jul	17 Jul	22 Jul
To	18 Apr	22 Apr	03 May	07 May	17 May	21 May	02 Jun	07 Jun	16 Jun	20 Jun	02 Jul	07 Jul	16 Jul	21 Jul	31 Jul
RA	13h 20m - 14h 15m		14h 15m - 16h 10m				16h 10m - 18h 05m				18h 10m - 20h 10m				
Block	3D	3DL	4L	4DE	4D	4DL	5L	5DE	5D	5DL	6L	6DE	6D	6DL	7L
Blk 1	P														
Blk 2															
Blk 3															
Blk 4															
Blk 5															
Blk 6															

Own Equipment (Enter name of instrument and give full description in *Discussion*)

Cass (with P3K):

Cass (without P3K):

Prime:

Semester 2018A Palomar Cover Sheet 200-in
Last updated: 27 July 2017 - RJB/AFB



OBSERVING REQUEST
Telescope Access Program, China

Year: 2018

Term: A

Proposal type: short-term

Virgo3D: Dynamics and Stellar Populations of Galaxies in the Virgo Cluster

P.I.: Zheng Zheng (NAOC; zz@bao.ac.cn;)

CoI(s): Eric Peng (PKU), Chengze Liu (SJTU), Sungsoon Lim (PKU), Elisa Toloba (UCB), Kyle Westfall (UCO/Lick), Patrick Cote (NRC), Raja Guhathakurta (UCSC), Laura Ferrarese (NRC), Chung-Pei Ma (UCB), Eric Emsellem (ESO), Alessandro Boselli (LAM), Joel Roediger (HIA), Ariane Lancon (University of Strasbourg), Adrien Guerou (ESO)

Abstract of Scientific Justification

Understanding the quenching of star formation is key to revealing the physical processes involved in galaxy evolution and transformation. Whether quenching is largely due to internal processes within galaxies or external processes due to the environment, and how these processes depend on the galaxy mass, are still open questions. These questions have been addressed through the study of the dynamics and stellar populations of relatively massive early type galaxies ($M_* > 10^9 M_\odot$) or the study of the Local Group dwarfs ($M_* < 10^7 M_\odot$), which has resulted in a morphology and mass gap in our understanding of these quenching processes, and about galaxy evolution in a more general context. Based on the deep Next Generation Virgo Cluster Survey (NGVS) imaging, we are conducting a 2-D spectroscopic survey of Virgo cluster galaxies that, for the first time, includes quenched and star forming galaxies across the full mass range $10^7 < M_* < 10^{12} M_\odot$. With this unique sample we will explore the relation between star forming and quenched galaxies as well as the environmental mechanisms that transform the former into the latter as a function of mass. Specifically, we will measure the angular momentum, dark matter content, and stellar populations across a large, unbiased, sample of galaxies in a mass range not yet explored, allowing comparisons to new cosmological simulations that are also now investigating this critical regime in mass. Our full survey targets a representative sample of galaxies in Virgo with $M_* > 10^7 M_\odot$. Here, we request 5 dark nights to observe 10 galaxies. This is part of a China-UC collaborative program to exploit the synergy between the sibling instruments Palomar/CWI and Keck/KCWI.

Summary of observing runs requested for this project

Run	Telescope	Cage	Instrument	PI	AO	Nights	Moon	Scheduling		Sharing	
								Optimal	Acceptable	Poss.	Adv.
1	P200		CWI			5	dark	Mar–Apr	Mar–Apr	yes	yes
or:											
1a										no	no

Scheduling constraints and unusable dates (up to 4 lines): We require dark nights, but given the relatively large requested block (5 nights), and the relative visibility of Virgo with respect to the Moon, we have identified some nights designated as gray that could be scheduled. They are: March 21-22, and April 19. We are willing to share half nights if there is a suitable program to be matched.

no text past this line

A * appended to the proposal type indicates a continuation proposal; a * appended to the name of a proposer indicates the proposer is a (graduate) student; a proposer whose name is underlined is certified on the proposed telescope/instrument combination; if a * appears within the PI or AO box in the observations summary table, the instrument is a PI instrument and/or Adaptive Optics are requested – signatures are required on the next page.

Target list (attach list if longer than 26 objects)				
#	Object	RA	Dec	mag / color / type / redshift / comment / etc.
1	VCC491	12:21:40.70	+11:30:6.12	$M_* = 10^{9.2} M_\odot$, $\log Re = 3.25$ pc, $\mu_{g,e} = 22.1$ mag/' ²
2	VCC938	12:26:46.68	+07:55:7.32	$M_* = 10^{9.5} M_\odot$, $\log Re = 3.24$ pc, $\mu_{g,e} = 22.5$ mag/' ²
3	VCC1049	12:27:54.84	+08:05:25.44	$M_* = 10^{8.9} M_\odot$, $\log Re = 2.78$ pc, $\mu_{g,e} = 22.5$ mag/' ²
4	VCC765	12:25:3.48	+13:14:40.92	$M_* = 10^{8.4} M_\odot$, $\log Re = 2.66$ pc, $\mu_{g,e} = 22.6$ mag/' ²
5	VCC1035	12:27:42.1	+12:05:23.28	$M_* = 10^{8.3} M_\odot$, $\log Re = 2.72$ pc, $\mu_{g,e} = 22.6$ mag/' ²
6	VCC672	12:23:53.47	+07:06:25.56	$M_* = 10^{9.4} M_\odot$, $\log Re = 2.91$ pc, $\mu_{g,e} = 21.7$ mag/' ²
7	VCC593	12:22:52.75	+06:40:46.92	$M_* = 10^{8.6} M_\odot$, $\log Re = 2.91$ pc, $\mu_{g,e} = 22.7$ mag/' ²
8	VCC787	12:25:18.14	+05:44:27.96	$M_* = 10^{8.9} M_\odot$, $\log Re = 3.15$ pc, $\mu_{g,e} = 22.5$ mag/' ²
9	VCC1410	12:25:18.14	+05:44:27.96	$M_* = 10^{8.9} M_\odot$, $\log Re = 2.97$ pc, $\mu_{g,e} = 22.4$ mag/' ²
10	VCC1348	12:31:15.72	+12:19:54.48	$M_* = 10^{8.8} M_\odot$, $\log Re = 2.74$ pc, $\mu_{g,e} = 22.6$ mag/' ²

If this program uses a PI instrument, attach the approval email from the PI to this proposal.

Graduate students (provide the following information for *each* student named as PI or CoI on the cover page.)

Student's Name	Advisor's Name	Thesis

Scientific Justification Describe the overall significance to astronomy and within the proposal's discipline. Limit text to one page, with a limit of two additional pages for figures, captions, and references.

Background context: A critical process in galaxy evolution is quenching, the process that stops star formation in galaxies. Understanding quenching in low mass galaxies is particularly important because modern cosmology suggests that galaxies form hierarchically (Λ CDM–cold dark matter), where low mass galaxies are the building blocks of more massive galaxies. Quenched low mass galaxies are always found close to a massive host while star forming low mass galaxies are typically found in isolation (Gavazzi et al. 2010, Geha et al. 2012). This suggests that the environment plays a key role in low mass galaxy quenching. It is not yet clear, however, at which mass range the internal (e.g., supernova, AGN) versus external (environmental) processes dominate quenching or even which environmental process (ram pressure stripping vs. harassment) dominates under certain conditions. In simulations, the recent Illustris cosmological simulation finds that the evolution of low mass galaxies is highly dependent on environment (Fig. 1; Mistani et al. 2015), making predictions for the observed properties of low mass galaxies that are a function of mass and environment. These and other studies suggest that an important transition takes place around a mass of 10^8 – $10^{10} M_\odot$, but it is exactly here where we have a dearth of data.

The key to future progress is to analyze unbiased samples of galaxies in a wide range of mass, internal properties, and located in regions of different density. While the high mass end ($M_* > 10^{9.5} M_\odot$) of the luminosity function has been extensively analyzed through dedicated 2-D spectroscopic surveys (SAURON, ATLAS^{3D}, MASSIVE, MaNGA, CALIFA, and SAMI), the low mass end ($M_* < 10^7 M_\odot$) is mainly based on the Local Group dwarf galaxies, and all surveys have put a special focus on the early-type population. This has created a ~ 2 dex mass gap, as well as a morphological gap, in our understanding of galaxy formation and evolution (Figure 1).

Our goal: We propose to use the unique integral field spectroscopic capabilities of P200/CWI and Keck/CWI to conduct a comprehensive survey of the stellar populations, dynamics, and dark matter content of an unbiased sample of star-forming and quenched galaxies in a wide range of mass ($10^7 < M_* < 10^{12} M_\odot$) that will fill the current mass gap. By measuring quantities such as specific angular momentum, stellar mass fraction, and the ages and chemical compositions of the stars, as a function of mass and environment, we can assemble a unified picture of galactic star formation and quenching (see the ATLAS^{3D} series of papers) for comparison to simulations. For this purpose, we need a representative and homogeneous catalog of nearby early and late galaxies down to low mass in environments of different density. The Virgo cluster, located at 16.5 Mpc (Mei et al. 2007), is the best location for this study for several reasons: (1) Virgo is so close that we can spatially resolve galaxies both photometrically and spectroscopically in the full range of masses proposed here; (2) Virgo is still under assembly, 300–400 star forming galaxies fall into the cluster every gigayear (Boselli et al. 2008a, Gavazzi et al. 2013), thus, it contains galaxies at all stages of evolution; (3) The local galaxy density varies strongly within the cluster, from very dense conditions in the core to being close to field density by its virial radius (e.g. Ferrarese et al. 2012); (4) The Next Generation Virgo Cluster Survey (NGVS) provides the deepest ($\mu_g \lesssim 29$ mag arcsec⁻²), most complete ($m_g \lesssim 26$), and homogeneous catalog of Virgo cluster galaxies that contiguously covers 104 deg²; and (5) NGVS optical data (u^*griz) is complemented with other multi-wavelength data (GALEX; Spitzer, Herschel, WISE; H I: ALFALFA; X-ray: ROSAT, Chandra, XMM) needed to address the environmental processes (e.g. Boselli et al. 2014).

The uniqueness of this survey: While large surveys have begun to include late-type galaxies (MaNGA, CALIFA, SAMI), none of them explore the mass range proposed here (their low mass limits are $10^{8.2}$ – $10^9 M_\odot$), or the Virgo cluster (all have a redshift cut that leaves Virgo out of their sample). Moreover, none of these surveys have the exquisite photometry of the NGVS, which allows us to combine spectroscopy and photometry of an unbiased, large sample of galaxies to study, as a function of mass, morphological type, structural properties, and environmental density: (1) angular momentum; (2) dark matter content; (3) stellar populations and star formation histories; (4) galaxy nuclei; (5) globular clusters; (6) kinematically decoupled cores; and (7) environmental processes like ram pressure stripping and harassment.

This proposal: This proposal is a continuation of our survey of ~ 30 galaxies in the mass range $10^8 < M_* < 10^{12} M_\odot$. In 2016A and 2017A, we observed a total of 14 galaxies. We plan to observe 10 more galaxies in 18A, 6 star-forming, and 4 quenched in the mass range $10^8 < M_* < 10^{10} M_\odot$.

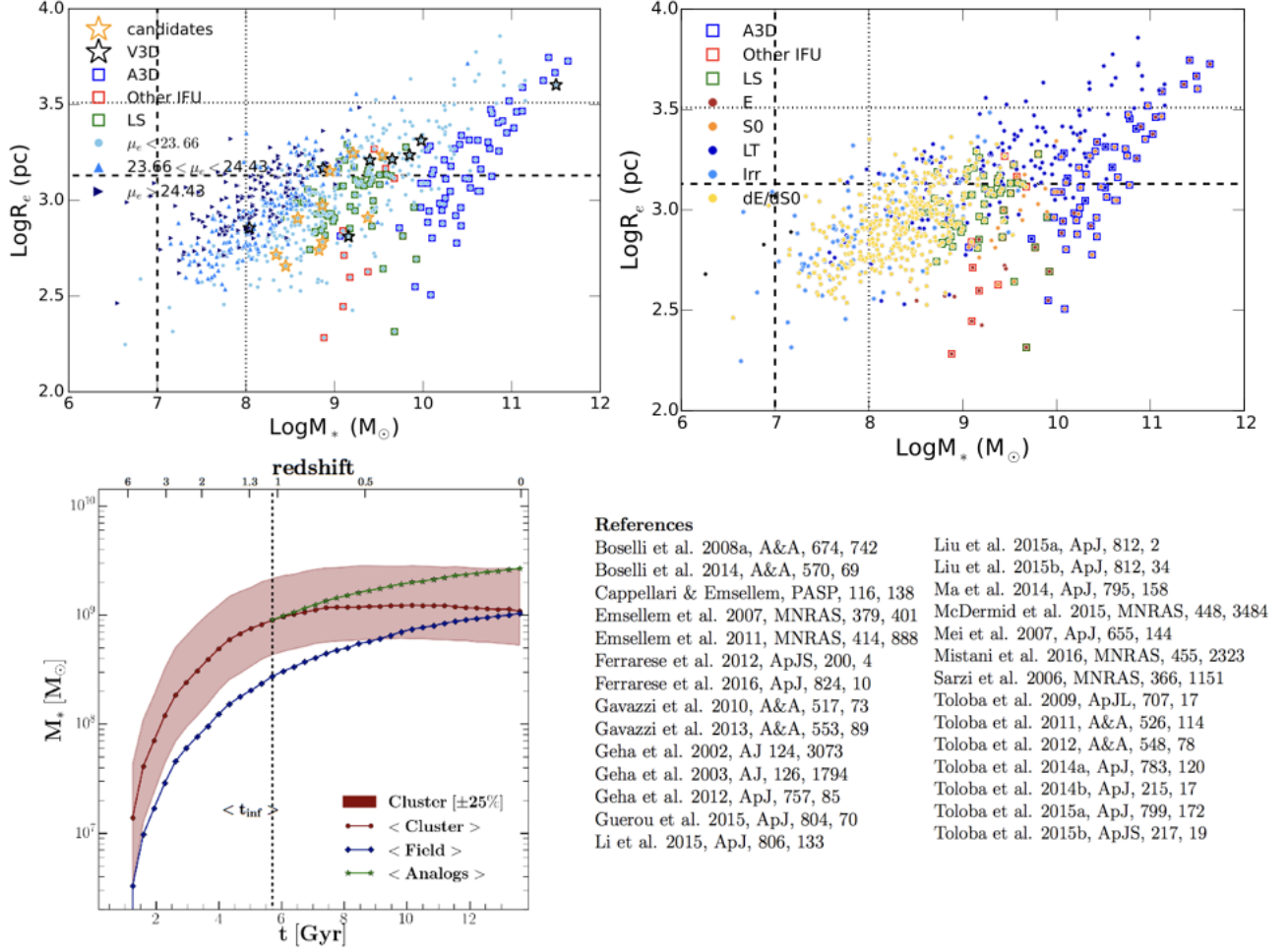


Figure 1: **Top left panel:** Mass-size relation for Virgo cluster galaxies. Different colors indicate the different morphologies. Symbols enclosed by a square indicate that they have various spectroscopic observations. The black stars indicate the observations from 2016A and the orange stars are our candidates in this proposal. The dotted and dashed horizontal lines indicate the size of the field of view for CWI and KCWI respectively. The dotted and dashed vertical lines indicate the range of masses best suited for CWI and KCWI, respectively. The sweet spot for CWI observations is therefore the region between the two horizontal lines and with masses larger than $10^8 M_\odot$. **Top right panel:** Same as the top left panel but color coded by the surface brightness at the half light radius R_e . The light and intermediate blue symbols are galaxies that can be observed in 2h without and with some spatial binning not larger than 10 spaxels, respectively. Dark blue triangles are galaxies that are better suited for Keck/KCWI because they need exposures longer than 4h with Palomar/CWI. These exposure times are the required ones to obtain a $S/N \sim 5$ at the half light radius and be able to measure radial velocities. **Lower panel:** Median stellar mass evolution for the sample of cluster (red) and field (blue) low mass galaxies simulated within Illustris (Mistani et al. 2016). Both samples have the same mass at the present time, but at earlier times, cluster low mass galaxies were more massive than their field counterparts. Both samples gain mass at a similar rate up to $z \sim 1$ when the growth of cluster low mass galaxies decelerates due to environmental effects, leading to lower overall stellar mass fractions and shorter star formation timescales. In contrast, a sample of low mass analogs that had the same properties as the progenitors of the cluster sample at the time of infall would outgrow in mass by up to a factor of 3 today (green line). We can test this by measuring the mass-to-light ratios and $[\alpha/\text{Fe}]$ for a large sample of low-mass galaxies.

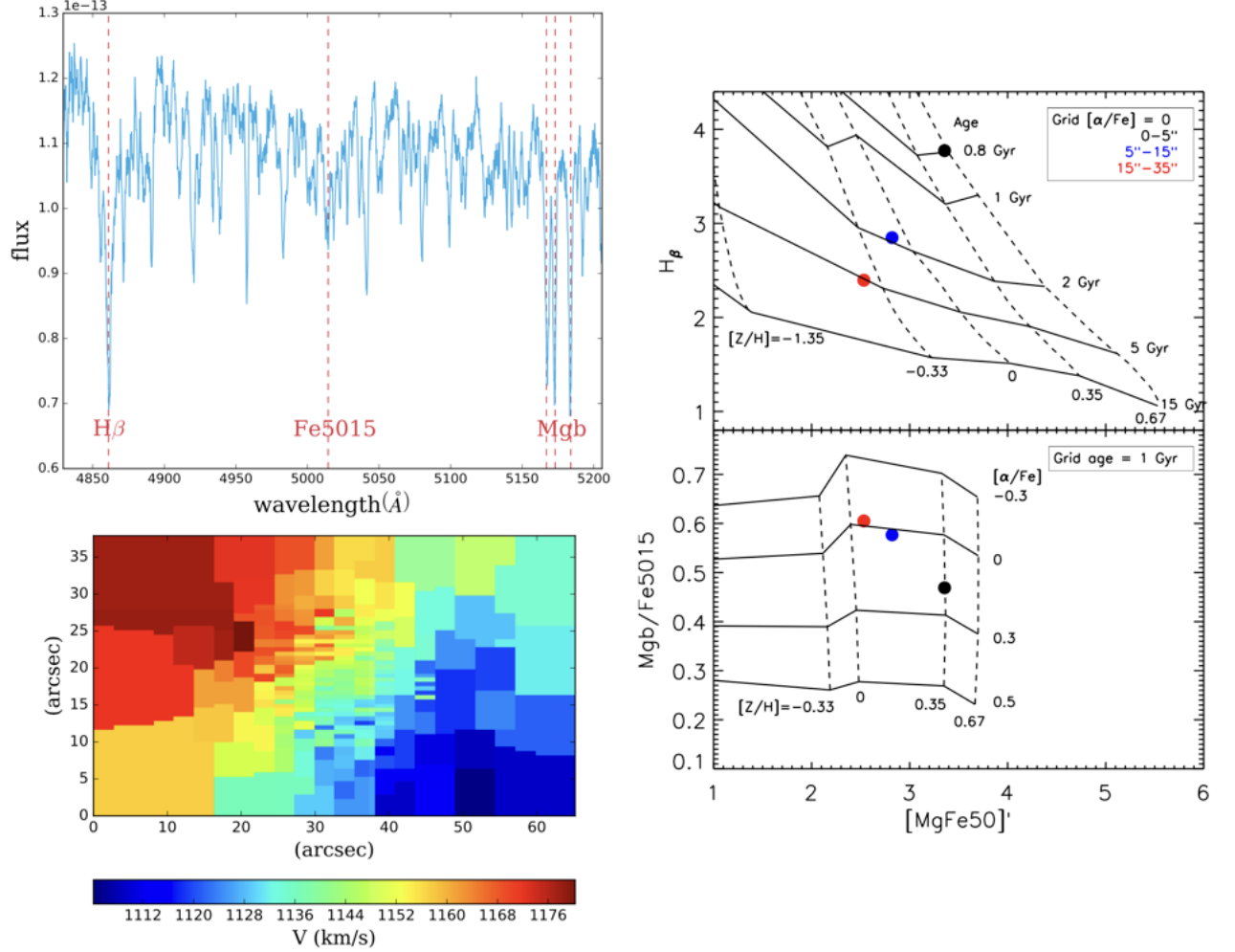


Figure 2: **Left (top)**: Integrated spectrum (de-redshifted) for VCC 1902 within one effective radius. The red dashed lines indicates absorption features we are interested in. **Left (bottom)**: The velocity field we measure as a result of Voronoi binning with $S/N = 30$. VCC 1902 is nearly face-on ($i \approx 20$ deg), but we can still discern a rotation of ~ 30 km/s. **Right**: Line index measurements plotted on the model grids of Maraston et al. showing clear, age, metallicity, and $[\alpha/Fe]$ gradients within the galaxy. These preliminary results show that our PCWI data are of high quality and can fulfill our science objectives. The strong gradients exhibited here are in contrast to the very weak gradients seen in more massive galaxies in MaNGA (Zheng et al. 2017).

Experimental Design Describe your overall observational program design. How will these observations contribute toward the accomplishment of the goals outlined in the science justification? Include information such as why the specific targets were selected, the sample size, the analysis plan, instrument choice, etc. Also briefly explain what expertise or effort each team member will contribute to the project. Describe any necessary non-standard calibrations (*up to one page*).

Instrumentation: Spectroscopic studies are traditionally based on long slits that cover a narrow region of the galaxy. If the dynamical structures, such as kinematically decoupled cores or kinematic twists, are not aligned with the slit, they are missed, and slits also integrate very little light in the galaxy outskirts. We plan to use the state-of-the-art integral field spectrograph Cosmic Web Imager (CWI) located at the 200'' telescope at Palomar.

Its intermediate resolution ($R \sim 5000$; instrumental dispersion $\sigma_{inst} \sim 26 \text{ km s}^{-1}$) is appropriate to resolve the stellar kinematics of galaxies in the full mass range proposed here (expected velocity dispersions of $15\text{--}300 \text{ km s}^{-1}$; with $S/N \sim 20$ it is possible to reach half the σ_{inst} ; Toloba et al. 2011). We will center the $\sim 375 \text{ \AA}$ spectral range at $\sim 5000 \text{ \AA}$ to observe $H\beta$, the Mgb triplet, and some Fe lines. This wavelength range, nearly 70% to the one used by the ATLAS^{3D} team but at significantly higher spectral resolution, has the prominent lines needed to measure stellar kinematics and stellar populations (Figure 2).

The sample: A mass complete survey is not feasible in Virgo due to its several hundreds of galaxies. We aim to target a representative sample of galaxies in the mass range $10^7 - 10^{11} M_{\odot}$. Together with observations from Keck/KCWI, we aim to observe ~ 50 galaxies, with half coming from each telescope.

We propose to target 10 galaxies in the mass range $10^8 - 10^{10} M_{\odot}$ (see Figs. 1, ??). Based on the observing strategies tested in 2016A, we will dither the integrations for the 4 galaxies with the smallest Re and get their sky from the blank regions in the edges of the images. For the remaining 6 larger galaxies, we will target blank sky regions close to the target that will be integrated for the same amount of time as the galaxy.

CWI and KCWI: Through UC, we have access to the Keck telescope where KCWI (Keck Cosmic Web Imager, sibling of CWI), which is now available. We have submitted a 2-night proposal to use KCWI for Virgo3D (PI: Westfall). We will exploit the CWI-KCWI synergy by observing the faintest and smallest galaxies (see Figure 1) with KCWI (smaller field of view, $20'' \times 15''$, but on a 10m telescope) and the brighter and larger galaxies, which do not fit in KCWI's FOV, with CWI.

Path from reduced data to science: Once the data are reduced using the CWI/KCWI pipeline, we will use the pPXF software (Capellari & Emsellem 2004) to obtain the radial velocity and velocity dispersion maps by applying a Voronoi binning technique when needed to increase the S/N . These measurements will provide cluster membership for those galaxies not yet spectroscopically confirmed, and the central spaxel provides the information for the nuclei if present. These measurements will be used to calculate the specific angular momentum λ_R , that combined with the flatness (ellipticity) will be used to classify the galaxies into slow and fast rotators (e.g. Emsellem et al. 2011; Toloba et al. 2015). Combining the dynamics and the light distribution we will perform dynamical modeling to investigate their dark matter content. The absorption spectral indices (e.g. $H\beta$, Fe5015, Mgb) will be used to analyze the stellar population and star formation efficiency from their ages, metallicities, and α enhancement following the same techniques as done by the ATLAS^{3D} team (McDermid et al. 2015). This analysis will be performed as a function of stellar mass, morphological type, star formation rate, structural properties, and environmental density.

Our team: The PI, ZHENG, is skilled in the analysis of galaxy IFU data (Zheng et al. 2017) through his work with MaNGA. He will lead the analysis of the PCWI data. Likewise TOLOBA and WESTFALL are experts in dwarf galaxies and IFU analysis. We conducted the NGVS photometric survey and are experts on dynamics and stellar populations (Emsellem et al. 2007, 2011, Geha et al. 2002, 2003, Toloba et al. 2009, 2011, 2012, 2014ab, 2015ab, Ma et al. 2014, Guerou et al. 2015, Li et al. 2015, Liu et al. 2015ab). A large fraction of existing spectroscopy in Virgo galaxies has been taken by members of our team. ZHENG and PENG will be responsible for overall project management, including observations, data analysis, and science. ZHENG, PENG, and TOLOBA will coordinate efforts for sample selection and observing preparations (CÔTÉ, MA, GUHATHAKURTA, ROCKOSI, LIU, SUN, LIM, BOSELLI), data analysis (EMSELLEM, MA, LIU, SUN, LIM, ROEDIGER, FERRARESE, LANCON, GUEROU), and writing of science papers.

Technical Justification *Justify the instrument configuration, the required signal-to-noise, exposure times. All requests for dark or gray time must include a detailed justification for the requested lunar phase. Specify the total time needed, and (if applicable) the minimum requested time (up to one page).*

Exposure times: Our goal is to cover the effective radius (R_e) of each galaxy with enough S/N to obtain a radial velocity with uncertainties of the order of 10–20 km s⁻¹ ($S/N \sim 5$; e.g. Toloba et al. 2011). With our experience in 2016A and 2017A we now have very accurate exposure time estimations. In 1h we can get $\mu_g \sim 22$ mag arcsec⁻² with a $S/N \sim 5$ per spaxel. In 2h we can reach $\mu_g \sim 22.6$ mag arcsec⁻² with the same S/N and in 4h ~ 23.0 mag arcsec⁻². Applying a Voronoi binning of 9 spaxels, these surface brightnesses go up to ~ 1 mag arcsec⁻² fainter for the same final S/N of 5.

Sky subtraction: We previously tested two techniques: (1) Offset skies with the galaxy completely out of the FOV; (2) Dithering, keeping the galaxy in the FOV, but moving it around. The nod-and-shuffle mode is unusable because it does not allow the needed wavelength coverage for the stellar population analysis. From our tests, we learned that dithering is only acceptable for the smallest galaxies ($R_e \ll 1$ kpc). We found that the best sky subtraction is obtained when the exposure sequence is galaxy–sky–galaxy and the exposure time of the blank sky region is the same as that for the galaxy.

The need for dark time: One of the reasons a survey of low mass galaxies has not yet been performed, is that this mass range is roughly where the galaxy surface brightness is fainter than the dark sky. This difficulty can now be overcome, in part, with the ability to integrate over many spaxels in modern, high-throughput IFUs. The relevant surface brightness in our sample galaxies is that at the effective radius. The dark sky surface brightness at Palomar¹ is $\mu_g = 22.2$ mag/arcsec² at the approximately same wavelength range. Thus, 9 of our 10 targets have $\mu_{g,e}$ fainter than the dark sky. In addition, we are largely observing stellar absorption features, which are sensitive to increases sky noise. Observing during gray time would increase the sky brightness by roughly 2 mag, which increases our exposure times by a factor of 6–7, which would make our survey prohibitively difficult to complete.

Analysis techniques: Higher S/N is needed for velocity dispersion and stellar population measurements based on absorption lines ($S/N > 15$; e.g., simulations by Toloba et al. 2011). We will take full advantage of CWI, that provides 1560 spaxels, and we will coadd them using the Voronoi technique to increase the S/N . In this way, our velocity dispersion and stellar population maps will have lower spatial resolution than the radial velocity maps. In Figure 2, we demonstrate our techniques using PCWI data of VCC 1902.

To study the stellar dynamics and stellar populations of star forming galaxies we will follow the same strategy as ATLAS^{3D}, which contains a few star forming galaxies in its sample. We will mask the emission lines present in our wavelength range to obtain the kinematic measurements and we will model and subtract them using the GANDALF software (Sarzi et al. 2006) for the stellar population analysis.

The pilot survey: In 2016A, we lost one night due to bad weather conditions. We observed 8 galaxies that include one very bright galaxy, M60 (for comparison to existing SAURON and MASSIVE data), and one extremely faint galaxy, VCC1523. Bright galaxies were exposed for 1–2 hours and faint galaxies for up to 5h. Fig. 2 shows the results for VCC 1902, an average galaxy in our sample. Prominent absorption lines such as H_β and Mgb are visible. Even though VCC 1902 is nearly face-on we can measure a rotation of ~ 20 km s⁻¹. This Figure shows that the survey proposed here is feasible with Palomar/CWI. This ambitious survey cannot be done in any other observatory, but for Palomar and Keck, because these are the only two observatories with large field-of-view and high resolution integral field spectrographs in the Northern hemisphere.

Total request: In February, March, and April, Virgo is visible for 7 hours per night at airmass above 1.6. To obtain the proposed sample of 10 galaxies, we request **5 full nights in dark time** (35 hours on Virgo). Our plan is to integrate for 1–3h per galaxy. Including the 50 % overheads for CWI, we will spend between 2–6h on each target. Low surface brightness absorption line spectroscopy in the optical range is only doable if there is no moon on the sky.

¹<http://lcogt.net/user/apickles/dev/Palomar/SkyBright/Aube.html>

TAP Usage and Context The TAC needs to understand the scope of this project —

(1) How many total TAP nights have already been allocated for this project (if any), how many are you requesting this semester, including any CFHT time from a linked proposal, and if more time will be necessary in future semesters, how many nights will you need to complete the project (best guess)?; (2) If an observing for this project comes from, or is being requested from, resources outside TAP, tell us about that observing, and how the TAP part fits in. (3) List any time from partner institutions, going toward (or being requested for) this project from TAP partners (e.g., Arizona, UC, Caltech, CFH, etc.), and who is the lead investigator on that time; (3) If you are collaborating with people who have access to other telescopes, especially if you are part of a large collaboration, tell us who is leading the project, and how TAP time and your participation fit in. 4) Please explain any ways in which this program would help fulfill the goals of the Telescope Access Program. (*up to one page*)

10 TAP nights have already been allocated for this project in 2016A and 2017A. Of these, 3 nights were lost due to bad weather conditions. We estimate that we will need 10 more nights at Palomar/CWI to complete the Palomar portion of the survey. We request 5 nights for semester 2018A.

These observations are part of a China-UC collaborative project that uses both Palomar/CWI and Keck/KCWI in complementary fashion. Z. Zheng and E. Peng are leading the China portion of the survey with access to CWI and postdoc involvement (S. Lim). Three UC staff or faculty—K. Westfall, P. Guhathakurta (UCSC), and C.-P. Ma (UCB)—are proposing for Keck/KCWI observing time in 2018A. We have already received KCWI commissioning data in 2017A to test our observing modes. The synergy between these two sibling instruments works well for this project: the larger and brighter objects that do not fit in the Keck/KCWI field-of-view (FOV) are best suited for Palomar/CWI, and the smallest and faintest galaxies are more suitable for a larger telescope with a higher resolution spectrograph like Keck/KCWI. We emphasize, however, the CWI program alone would extend the mass range of observed galaxies to new and interesting limits, and would enable plenty of new science.

This project is a very balanced collaboration. Project management, data acquisition, and analysis will be led by Zheng and Peng and their research groups at NAOC and PKU, and by K. Westfall (UCO) and E. Toloba (U. of the Pacific) in California. Canadian (Côté, Ferrarese, Roediger) and European (Emsellem, Boselli, Lançon, Guerou) co-Is are part of the NGVS team, and will provide critical photometric and spectroscopic support. In effect, TAP Palomar time will be used to gain access to many nights of Keck time over the next few years.

Previous Use of TAP Facilities

List **all** allocations of telescope time for the present project and allocations for other projects on facilities available through TAP during the past 2 years, together with the amount of time awarded (e.g. 2 nights, 6 hours), the percentage of this that was useful (e.g. 80), and text describing the current status of any data obtained. Cite publications that were based (to a significant extent) on data obtained at TAP facilities. Mark those allocations related to the present proposal (i.e. precede text with `\related` command). (**up to one page**)

- ★ 2015A-13, PI: Zheng Zheng, “Probing physical properties and formation mechanisms in outer disks of nearby disk galaxies using deep SWIFT IFS”, 4 nights, P200/SWIFT. The observation was not successful because of bad weather and instrument failure.
- ★ 2015B-12, PI: Zheng Zheng, “Spectroscopy of a strong lens system with a spectacular Einstein ring,” 1 night, P200/DBSP. Only one of the targets (50%) is observed because of bad weather. The observed data has been reduced and a paper has been published in 2017 (Wang et al. 2017, MNRAS, 468. 3757).
- ★ 2015B-33, co-I: Zheng Zheng, “Probing the structure of dark matter haloes by imaging spectacular Einstein rings?”, 1.5hrs, CFHT MegaCam, B queue. All the targets were observed and have been fully reduced. This is an adjunct proposal with the one listed above. A paper using these data has been published (Wang et al. 2017, MNRAS, 468. 3757).
- ★ 2017A-05, co-I: Eric Peng, “Dynamics and Stellar Populations of Galaxies in the Virgo Cluster”, P200/CWI, 5 nights, 70% useful. The second run for Virgo3D, we observed 7 galaxies in variable weather conditions.
- ★ 2016A-02, co-I: Eric Peng, “Dynamics and Stellar Populations of Galaxies in the Virgo Cluster”, P200/CWI, 5 nights, 70% useful. The first run for Virgo3D, we observed 8 galaxies, trying a variety of observing sequences and sky subtraction strategies. We are now familiar with the CWI pipeline, can make exposure time calculations based on real data, and have a well-understood sky subtraction procedure.